

Exhaust, no inflow

Remaining room $K - N(t)$ is spent and never returned. The integrated rate $\mathcal{B}(t)$ converges, and S_t has a proper limit law.

logistic speciation

Exhaust, with inflow

Potential V is consumed per capita and restocked at rate ϕ . The process cycles rather than settling.

rise-run-ruin-rejuvenation; vitality

Moving critical point

Resistance update drags the state of no preferred direction after the leader, and so restores criticality.

Pimentel competition

Rupture as a join

The intracellular phase ends at rupture, rate $\delta_1 X_t$. The burst-size law integrates that time out, so the phases compose.

two phases of infection